Master's Thesis Proposal

Harvard Master's in Data Science

Peter Flo Grindehollevik
pgrindehollevik@g.harvard.edu

Submission: April 10, 2026

1. Title

Attention-Guided Router Steering: A Feature-Learning Lens on Behavioral Intervention in Sparse Mixture-of-Experts Language Models.

2. Advisor

Adityanarayanan (Adit) Radhakrishnan — Assistant Professor, MIT Department of Mathematics; Associate Member, Broad Institute of MIT and Harvard (aditradha.com).

3. Abstract

Sparse mixture-of-experts (MoE) language models such as Mixtral 8x7B route every token to a small subset of feed-forward experts via a learned router. Recent work (SteerMoE [21]) proposes that this router is itself a viable intervention site: contrastive prompt pairs identify experts whose activation rate differs in the presence of a target concept, and adding a bias to those experts' router logits at decode time can steer generation toward the concept. In a faithful re-implementation I find that the recipe transfers weakly to Mixtral under a question-only concept-transfer test, and that the underlying signal is diffuse: the top twenty (layer, expert) cells out of $32 \times 8 = 256$ capture less than ten percent of the total $|\Delta_{\ell,e}|$ mass. This thesis proposes **Attention-Guided Router Steering (AGRS)**, a drop-in replacement for SteerMoE in which the per-token contribution to the readout is weighted by the attention mass that token sends toward a concept-bearing prefix span. AGRS reduces to SteerMoE when the attention weights are uniform, which isolates the contribution to the readout step. I evaluate AGRS against SteerMoE and three hidden-state steering baselines across four MoE checkpoints and four concept families, with a pre-registered decision gate before the more expensive generalization runs. The mechanistic chapter uses the Average Gradient Outer Product (AGOP) framework from my advisor's recent work to characterize the concept-relevant input subspace each selected expert responds to, and asks whether AGRS chooses better-aligned experts than SteerMoE does.

4. Motivation and Background

Prior research. My prior project at Harvard was on hyperbolic prototype embeddings for hierarchical artistic style on WikiArt,[1] which trained matched-dimension Euclidean and Poincaré-ball prototype classifiers across 150 seed-replicated runs and reported a robust local-retrieval advantage for the hyperbolic geometry. That project taught me how to design a tightly scoped representation-learning study, run an honest mixed-results comparison, and ship a reproducible end-to-end artifact — skills directly transferable to this thesis.

Preliminary results. For the past several months I have been running activation-steering experiments on Mixtral 8x7B-Instruct on the MIT ORCD cluster, and a working draft of the conference paper that came out of those experiments is checked into the project repository.[2] The empirical hook for the thesis is concrete. Across 25 question-only test cases on Mixtral, swept across eight (α, k_+, k_-) SteerMoE configurations, three things hold consistently: (i) the concept-token hit rate moves by at most 0.04 between baseline and SteerMoE; (ii) response length contracts monotonically with the steering coefficient (-2.16 to -5.52 words on average); and (iii) the per-(layer, expert) risk-difference signal is diffuse, with the top twenty cells out of 256 capturing less than 10% of the total $|\Delta|$ mass. If the readout is diffuse and the intervention barely moves the model, then reweighting the readout by attention to the concept span is the lightest-touch method change that could plausibly fix it.

Why this matters. Behavioral steering at inference time is one of the few corners of mechanistic interpretability where small interventions on frozen models produce measurable, causal behavioral effects. Most published steering methods — ITI [9], Activation Addition [20], Representation Engineering [23], Contrastive Activation Addition [14] — operate on the residual stream of dense transformers. Sparse MoE models give us a structurally different intervention site: the router logit, whose causal effect is mediated through a discrete selection over a small expert pool. Whether the readout-and-intervene paradigm transfers from dense to sparse architectures, and whether it does so with the *same* readout rules, is an open empirical question with direct deployment relevance — a large fraction of the open-weight frontier is now MoE [7, 13].

Connection to my advisor's research. Adit's recent work introduces the **Average Gradient Outer Product (AGOP)** $M = \mathbb{E}_x[\nabla_x f(x) \nabla_x f(x)^\top]$ as a mathematical mechanism by which neural networks acquire features [15, 3, 16]. The AGOP encodes *which input directions a model treats as informative for a given prediction*. For sparse MoE routing the analogous question is *which input directions a router's gating function for expert e at layer ℓ treats as informative*, and the mechanistic chapter tests whether AGRS picks experts whose top AGOP eigenvector aligns with concept-prototype directions in a way SteerMoE does not. This is the specific intellectual bridge to my advisor's existing toolkit.

Future plans. After this master's I intend to pursue research roles in AI interpretability and behavioral evaluation of large language models, with industry research labs focused on alignment and post-training intervention methods as the primary target. The thesis is sized to produce a publication

[1] github.com/pgrindehollevik-harvard/hyperbolic.

[2] Working draft and code: github.com/pgrindehollevik-harvard/MoE_attention_guided_steering (paper-submission branch).

artifact (a NeurIPS-tier method paper) and the cluster-scale empirical-research experience that those programs evaluate against.

5. Goals and Expected Outcomes

1. **Method.** Define and implement AGRS as a drop-in replacement for SteerMoE on Mixtral 8x7B and OLMoE 1B-7B, with unit tests including the property test that AGRS reduces to SteerMoE when attention weights are uniform. *Success criterion:* AGRS produces strictly more-concentrated risk-difference tables than SteerMoE on $\geq 4/5$ pilot fear concepts, measured by the top-20 share of $|\Delta|$ mass.

2. **Empirical evaluation.** A pre-registered comparison of AGRS against five alternatives — unsteered baseline, SteerMoE, ITI [9], Activation Addition [20], and Representation Engineering difference-of-means [23] — across four MoE checkpoints (Mixtral 8x7B [7], OLMoE 1B-7B [13], Qwen2-MoE 14B [22], DeepSeek-MoE [5] if licensing permits) and four concept families (fears, professions, refusal targets, sentiment polarity). Sample size $n = 250$ cases per condition, five seeds. Primary metric: concept-transfer rate (CTR) scored by an LLM judge cross-validated against a 200-case human-rater subset. *Success criterion:* AGRS lifts CTR by ≥ 0.10 over SteerMoE on Mixtral / fears with non-overlapping paired bootstrap 95% CIs, and produces a positive CTR shift in $\geq 3/4$ models and $\geq 3/4$ concept families.

3. **Mechanistic analysis with AGOP.** Estimate the AGOP of selected experts' gating functions over the contrastive prompt pool, and report the cosine alignment between the top AGOP eigenvector and the concept-prototype embedding direction. *Success criterion:* AGRS-selected experts have systematically higher alignment than SteerMoE-selected experts on Mixtral / fears; the per-cell (model, concept) correlation between alignment and CTR gain reaches $|r| \geq 0.5$.

Deliverables. A submitted NeurIPS or ICLR main-track paper, a thesis monograph, and an anonymized public artifact tree under CC-BY (executed notebooks, generation metadata, qualitative reviews, activation tables, steering plans, judge transcripts, Apptainer image, and OSF pre-registration document).

6. Scientific Justification

Three things make this contribution scientifically interesting and innovative beyond a standard methods paper.

First, the *readout vs intervention* distinction is rarely isolated in the activation-steering literature: most published methods bundle a readout choice with an intervention site, making it hard to tell which is doing the work. AGRS reduces to SteerMoE under uniform attention. The same intervention site (router logits), the same generation pipeline, the same experiment harness — only the readout changes. This makes the contribution isolable in a way the literature has not previously managed for MoE steering.

Second, the diffuse-signal finding on Mixtral is a structural observation about *when* routing-bias steering should and should not work. The AGOP mechanistic chapter turns the diffuse observation into a prediction: routing interventions succeed where selected experts are aligned with concept-relevant input directions, and fail where they are not. Either AGRS clears the gate and the prediction is validated, or it does not and the diffuse-signal finding becomes a structural limit on how well any router-bias method can be expected to work on this class of model.

Third, the empirical rigor commitments — pre-registration on OSF before the gate-determining experiment runs, paired bootstrap CIs across seeds, judge-vs-human cross-validation, full artifact release under CC-BY — are not standard in this corner of the literature. They are standard at the venues this thesis targets, and they raise the bar in a direction the field benefits from.

7. Literature Review

The thesis sits at the intersection of four literatures.

(i) Activation steering on dense transformers. The line begins with latent steering vectors [17] and matures into Inference-Time Intervention [9], which adds a learned direction to a small set of attention heads to reduce hallucination; Representation Engineering [23], which formalises the readout-and-edit problem as a contrastive operation on the residual stream; Activation Addition [20], which projects the contrastive signal across multiple residual layers; and Contrastive Activation Addition [14], which extends the difference-of-means readout to Llama 2 with careful per-layer reporting. Tan et al. [18] audit the generalization and reliability of these steering vectors and find that small readout choices substantially affect downstream behavior. A parallel line uses sparse autoencoders to extract monosemantic feature directions [4, 19, 12] and uses linear probes to read concept-relevant directions [1, 11]. On the safety side, Arditi et al. [2] showed that refusal in Llama 2 is mediated by a single direction, and Lee et al. [8] gave a mechanistic account of how alignment algorithms reshape steering directions. None of these methods target sparse MoE routing.

(ii) Sparse Mixture-of-Experts language models. Mixtral 8x7B [7] and OLMoE 1B-7B [13] released the open-weight checkpoints this thesis builds on. Qwen2-MoE [22] and DeepSeek-MoE [5] round out the cross-architecture set. Each sparse layer routes every token to a small subset of experts via a learned router; the discrete routing decision is the structure behavioral steering exploits. The mechanistic-interpretability literature on routing itself is younger: Dai et al. [5] reports that DeepSeek-MoE's fine-grained routing achieves stronger expert specialization. Routing-pool size and routing density are axes we expect AGRS's gain to vary along.

(iii) Steering sparse MoE models. SteerMoE [21] is the upstream method this thesis builds on and amends. The recipe collects per-token router logits on contrastive prompt pairs, computes a risk-difference table over $(\text{layer}, \text{expert})$ cells, globally selects the strongest positive and negative experts, and adds a fixed bias to those experts' router logits at decode time. The readout is unweighted averaging over a target span. The diagnostic results in my preliminary draft show this readout produces a diffuse risk-difference signal on Mixtral and correspondingly small behavioral shifts under a question-only test. Adjacent work on MoE compression and expert pruning [6] treats experts as the unit of intervention but does not target concept transfer at decode time.

(iv) Feature learning, AGOP, and Recursive Feature Machines. The advisor's recent work [15, 3, 16] introduces the AGOP as a mathematical mechanism through which neural networks acquire features. The AGOP encodes which input directions a network treats as informative for a given prediction; the Recursive Feature Machine (RFM) family gives recovery guarantees in the linear case. The Tuned Lens [10] provides a cross-layer probing view of similar feature questions. The mechanistic chapter of the thesis uses AGOP to characterize which input directions a Mixtral expert's gating function is sensitive to, and asks whether AGRS's selections align with those directions in a way SteerMoE's do not.

Where this thesis fits. The gap is identifiable. The attention-guided readout machinery exists for dense-transformer steering (thread i) but has not been ported to MoE routing (thread iii). MoE checkpoints (thread ii) are the natural substrate where the readout question matters most — the diffuse-signal finding on Mixtral demonstrates this. And the feature-learning toolkit (thread iv) has not been applied to characterize routing-decision sensitivity. The thesis closes all three gaps with one method (AGRS), one evaluation, and one mechanistic analysis.

8. Methodology

The project is mainly computational. Every claim is empirical and reproducible from a pinned codebase, a pinned environment, and a fixed set of model checkpoints.

The AGRS readout, formally. For a contrastive prompt pair (p^+, p^-) with concept-bearing prefix span $\mathcal{C} \subset p^+$, let $R_{\ell,t,e} \in \{0,1\}$ indicate that expert e is selected by Mixtral top-2 routing at layer ℓ, position t; let $a_t = \max_h \sum_{t' \in \mathcal{C}} A^{\ell^\star}_{h,t\to t'}$ be the per-token attention mass to the concept span at a chosen reference layer $\ell^\star$ (the simplest default is the last layer; the chapter ablates this choice). The AGRS attention-weighted activation rate on p^+ is

$$r^+_{\ell,e} \;=\; \frac{\sum_t a^+_t \cdot R^+_{\ell,t,e}}{\sum_t a^+_t}, \qquad \Delta_{\ell,e} \;=\; r^+_{\ell,e} - r^-_{\ell,e}$$

with $a^-_t \equiv 1$ on the unprefixed prompt. The decode-time intervention is identical to SteerMoE: add $\pm\alpha$ to the top-k_+ positive and top-k_- negative experts' router logits at every forward pass during generation. SteerMoE is recovered exactly when $a^+_t \equiv 1$.

Computational pipeline. Per-token router logits and attention weights are captured via Hugging Face forward hooks, sliced to the shared statement-body target span, aggregated into the risk-difference table, used to build a sparse expert-activation plan, and re-injected at decode time. Generations are scored by an LLM judge (GPT-4o as primary, Claude as cross-check) on a 0–2 ordinal scale for thematic concept transfer, with a human-rater subset of 200 cases for judge calibration (Cohen's κ, Spearman). Statistical reporting is paired bootstrap 95% CIs across seeds and paired t-tests on seed-collapsed deltas.

Mechanistic AGOP analysis. For each selected expert e at layer ℓ, treat the expert's gating function $g_{\ell,e}(x)$ as a real-valued map from token-position embeddings to the pre-routing logit, and estimate

$$M_{\ell,e} \;=\; \mathbb{E}_{x\sim p^+\cup p^-}\Big[\nabla_x g_{\ell,e}(x)\, \nabla_x g_{\ell,e}(x)^\top\Big]$$

over the contrastive prompt pool. Report cosine alignment between the top eigenvector of $M_{\ell,e}$ and the concept-prototype embedding direction, per selected expert. The empirical question is whether AGRS picks better-aligned experts than SteerMoE.

Models and compute. Mixtral 8x7B-Instruct-v0.1 (8-bit with `bitsandbytes` CPU offload), OLMoE 1B-7B-0125-Instruct, Qwen2-MoE 14B-A2.7B, and DeepSeek-MoE-Lite if licensing allows; total budget $\sim$ 305 GPU-hours on two NVIDIA A100s per generation job on the MIT ORCD `mit_normal_gpu` partition across the fall and spring.

Reproducibility. Every experimental step is a Jupyter notebook with a `papermill` parameters cell, run identically on a laptop in smoke mode and on the MIT ORCD cluster via a paired `sbatch` wrapper that calls `papermill` headlessly. All generation calls are at temperature 0 with explicit seeds; every model load asserts a recorded Hugging Face revision SHA; every run writes a `provenance.json` with the git SHA, cluster job id, and wallclock. The cluster runtime is a single Apptainer image. The AGRS hypothesis is pre-registered on OSF with the decision gate before the gate-determining experiment runs. The full `experiments/` tree (executed notebooks, generation metadata, judge transcripts) ships at submission under CC-BY.

Timeline.

Months	Milestone
May–Jun 2026	Reproducibility hardening; `make paper` regenerates draft on clean checkout.
Jun–Jul 2026	Evaluation harness; judge–human Spearman ≥ 0.7 on 200-case subset.
Jul 2026	AGRS implementation; unit tests; OSF pre-registration filed.
Aug 2026	Pilot run on Mixtral / fears; decision gate evaluated.
Sep–Oct 2026	Full six-method comparison on Mixtral / fears.
Nov 2026	Cross-model generalization (Mixtral, OLMoE, Qwen2-MoE, DeepSeek-MoE).
Dec 2026	Cross-concept generalization (fears, professions, refusal, sentiment).
Jan 2027	AGOP mechanistic analysis; alignment-vs-CTR scatter.
Feb–Mar 2027	Statistics, thesis monograph, conference submission.

Risks and pre-declared pivots. If the August pilot fails the pre-registered ΔCTR $\geq +0.10$ threshold on Mixtral / fears, the thesis pivots in one of two registered directions: (a) the same attention-weighted readout is applied to a hidden-state intervention site, isolating the readout from the intervention site; or (b) the thesis becomes a focused diagnostic-and-AGOP study of why the routing signal is diffuse on Mixtral. Either pivot is described in the OSF pre-registration so it cannot be backfit.

Companion materials (`paper-submission` branch): working draft of the conference paper at `paper/docs/paper/main.pdf`, the phased revision plan with code-to-build manifest at `paper/docs/REVISION_PLAN` the notebook-as-pipeline scaffolding at `notebooks/README.md`, and the projected final-paper shape at `paper/docs/EXPECTED_FINAL_PAPER.md`.

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